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By

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Title:

Predicting Lung Cancer Incidence from CT Imagery

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Abstract

Lung cancer is one of the most common forms of cancer worldwide, and is responsible for a large number of deaths and significant health care costs. Human radiologists typically use low-dose computed tomography scans of patients lungs to assess an individual’s risk of lung cancer, usually by inspecting the images for the presence of tissue growths called “nodules” that are a common precursor to cancer. However, even for highly trained radiologists, detecting nodules and predicting their relationship to cancer are challenging tasks, leading to both false positive and false negative results that can adversely affect patient health.

Our initial objective in this project was to determine whether we could make direct inferences about a patient’s risk of lung cancer based on the application of deep learning models to “raw” CT imagery, but our attempts to train a three-dimensional convolutional neural network over full lung images were largely unsuccessful, perhaps due to the low signal-to-noise ratio that also makes human diagnosis difficult and and the lack and imbalance of the dataset for training is also a reason. We used 769 lung CT scans in this project, and the data used was pre-processed to ensure all the data used was in the isomorphic form, which are suitable for training deep neural network. The trained and tested model was assessed resulting in quantitative measures such as accuracy, sensitivity (recall), specificity, precision, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC).

The performance was evaluated at 68% accuracy of the test data to the built model. Still, there is so much more that can be improved. This approach provides a look into the the advantages and disadvantages of three-dimensional neural network, the future of artificial intelligence and how these deep learning models may help doctors in diagnosing lung cancer in the future.

Keywords: Artificial intelligence, medical imaging, diagnosis, lung cancer, CT scans, deep learning, three-dimensional convolutional neural network, data imbalance.

Contents

1	Introduction	10
1.1	Background and Context	10
1.2	Problem Statement	11
1.3	Research Objectives	12
1.4	Significance of the Study	12
1.5	Thesis Structure	13
2	Literature Review	14
2.1	Lung Cancer and Screening Methods	14
2.1.1	Lung Cancer Overview	14
2.1.2	Current Screening Methods	15
2.2	CT Scan Technology Overview	15
2.3	Application in Medical Imaging	16
2.3.1	Application of Deep Learning in Medical Imaging	16
2.3.2	Application of 3D CNN in Medical Imaging	16
2.4	Related Works	17
3	Material and Methodology	18
3.1	Material	18
3.2	Methodology	22
3.2.1	Overall Framework	22
3.2.2	Pre-Processing Data	22
3.2.3	Model Configuration and Training	27

3.2.4	Evaluation Metrics	30
3.2.5	Tools and Libraries	32
4	Results and Discussion	33
4.1	Results	33
4.2	Discussion	34
5	Conclusions, Limitations and Future Works	36
5.1	Conclusions	36
5.2	Limitations	38
5.3	Future Works	39

Acronyms

3D CNN Three-Dimensional Convolutional Neural Network. 10

AI Artificial Intelligence. 10

AUC-ROC Area Under the Receiver Operating Characteristic Curve. 12

CNN Convolutional Neural Network. 10

CT Computed Tomography. 10

DL Deep Learning. 10

LDCT Low-Dose Computed Tomography. 15

ML Machine Learning. 10

MRI Magnetic Resonance Imaging. 15

NCI National Cancer Institute. 18

NLST National Lung Screening Trial. 18

PET Positron Emission Tomography. 15

WHO World Health Organization. 10

List of Figures

3.1	Number of slices per patient	19
3.2	Number of patients by label	19
3.3	Lung CT scans	20
3.4	Overall framework	22
3.5	3D CNN model architecture	28

List of Tables

3.1	Important attributes used in the study	21
3.2	A sample of the labels.csv file	21
3.3	Substances and their corresponding Hounsfield Units	24
3.4	3D CNN model architecture	28
3.5	Hyper-parameters	29
3.6	Tools and libraries	32
4.1	Model performance	33

List of Algorithms

1	The proposed pseudocode for load data and add missing metadata . .	23
2	The proposed pseudocode for convert raw pixel to HU	25
3	The proposed pseudocode for resample	26
4	The proposed pseudocode for normalization	27

Chapter 1

Introduction

1.1 Background and Context

Lung cancer continues to be not only a major threat to human life, but also ranks as the leading type of cancer in terms of incidence and death. According to the World Health Organization (WHO) global report prepared in 2020, there were about 2.2 million new cases of lung cancer together with 1.8 million of lung cancer deaths. Lung cancer is also in the list of the most frequent cancers in Vietnam and the incidence rate is relatively higher in men and it contributes a lot of load on the health and societal system.[1]

It is therefore recommended that lung cancer be diagnosed at an early stage so that the probability of the patient surviving can be boosted. Currently, computed tomography or also referred as Computed Tomography (CT) scan is now the most effective technique in imaging of lung carcinoma. It provides the picture of internal structure of the lungs and helps the doctors identify the cancerous growth in the form of suspicious lesions. But considering a significant amount of CT scan images for reading and analysis is quite skillful and might be time consuming and is expensive when the medical facility is not well endowed.

Nowadays with the intensive development of Deep Learning (DL) technology, including into one of the subdivisions of Artificial Intelligence (AI), gave a new hope to diagnose in the sphere of medicine. Convolutional Neural Network (CNN) in particular have proved that deep learning models indeed surpass traditional Machine Learning (ML) algorithms in terms of accuracy in analysis of medical images and automatic feature extraction to classify lesion types.

In this regard, this project is aimed at realizing the potential of Three-Dimensional Convolutional Neural Network (3D CNN) model for constructing lung cancer prediction models based on CT scan. Thus, these models makes it easier for doctors, who already receive a large number of requests, and can positively affect the diagnostic work both in terms of increasing the accuracy, as well as the speed of the results, including in cases with small, barely visible lesions.

We hope that this project shall help in achievement of the application of artificial intelligence in the medical practice particularly in diagnosis of lung cancer. Thus, the proposed project with the link of medical knowledge and information technologies will create a powerful tool for doctors, which will help to increase the survival rate of patients.

1.2 Problem Statement

While CT technology plays a vital role in the discovery of lung cancer in the early stages accurately judging numerous scans proves difficult. This requires experienced specialists who take considerable time to complete their duties amid high workloads or when they lack skills. This not only burdens medical facilities but can cause misdiagnoses such as inaccuracies in assessments or missed small lesions that disrupt treatment results and progress.

To tackle the mentioned challenges effectively, this project intends to build a lung cancer prediction system that applies the three-dimensional convolutional neural network for analyzing patient CT slices directly. The goal of this is to create an effective resource for doctors to quickly identify and classify lung cancer in the early phases.

Nevertheless, construction of an efficient and dependable system of lung cancer prediction is rather a challenge. This project will face several key challenges, including:

Data: The approach of having a "enough" quantity and good quality of precisely labeled CT scan data is fundamental when developing a deep learning model. However, this may not be easily feasible especially as patient information security matters arise and the special high level of technicality in labeling. In this case, the dataset we used was imbalanced, so keep in mind that it could cause several problems in this thesis later.

Model optimization: Despite the fact that the proposed model retains the ability to train three-dimensional spatial features from CT scan data, its architecture is quite complex in nature and the computational burden is quite high. One of the main problems when tuning the model is to maximize its performance while maintaining a high throughput rate.

Evaluation of performance: The process of assessing the model implies strict criteria and procedures that regard the accuracy and sensitivity of the system, as well as to the specificity of the used prediction formula in a realistic context.

This project will aim at solving the above challenges, or at least try to solve some of them, understand, and implement higher levels of deep learning for enhancing the lung cancer prediction model. Thus, by solving these difficulties, we hope it will make a significant contribution to medical diagnosis and, thus, to the fight against lung cancer by doctors.

1.3 Research Objectives

Considering the issues stated above and about the chosen three-dimensional neural network model application, this project outlines certain research objectives that are directed to three main areas.

First, the project is on creating and training a 3D CNN model to solve the predict incidence of lung cancer from CT scan images. This process also entails identifying and choosing appropriate 3D CNN architecture to be used in the model besides presenting some pre-processing methods to enhance the model learning capability.

Subsequently, the proposed 3D CNN model will be assessed according to effective measures like accuracy, sensitivity, specificity, precision, F1-score and Area Under the Receiver Operating Characteristic Curve (AUC-ROC). Then the results of the model will be further discussed and compared to prior work employing 3D CNN models or other designing methodologies for predict incidence of lung cancer from CT scans.

The final significant contribution of the project concerns the role of the project as the indispensable aid for doctors when it comes to the identification of lung cancer, especially its early stages to enhance the diagnosis and treatment processes. This model also liberates doctors from carrying out many tasks, and enhances their performance in arriving at better decisions on patient treatments. The project also advances the use of artificial intelligence in medicine by also proving the efficiency of 3D CNN models in predicting lung cancer. Thus, the objectives established in the present project will make it possible to offer an effective solution and help enhance the quality of healthcare with a view to increasing the prognosis of patients.

1.4 Significance of the Study

This research makes substantial contributions to the development of artificial intelligence in the assessment of disease symptoms with particular reference to lung cancer at its early stage. Besides, this project opens up a new way of segmenting CT scan images of lung cancer and supports the conjecture that the 3D CNN model can process medical data. The success of this research will bring practical benefits to the community:

Supporting doctors: This thesis proposes the automatic lung cancer prediction model; it will serve as a supporting tool for doctors, which will reduce doctors' work and increase the likelihood and speed at which a diagnosis is made. This brings good health benefits particularly where resources are scarce in a health sector, the doctor is relieved of quickly solving simple cases and can make appropriate decisions regarding patients treatment.

Early detection of lung cancer: The "small and obscure" lesion discovery of

the 3 dimensional neural network model allows lung cancer to be detected when the probability for treatment is higher. This has a very important role in changing the treatment outcome and increasing the likelihood of the patient to survive.

Cost savings: An application of image analysis of CT scan has the potential of lowering the cost of the diagnosis and treatment of lung cancer thereby creating savings to the patients, the health system, and the economy as a whole.

Promote research and development: It is expected that this study will make an input towards the growth of the use of artificial intelligence in healthcare, presenting the foundation for a further study of lung cancer diagnoses and therapy as well as different diseases.

In other words, the presented work holds considerable scientific and practical relevance since it can satisfy the needs of the academic community and enhance the quality of health care by helping doctors save more lung cancer patients.

1.5 Thesis Structure

In this section, we will present the overall structure of this thesis, including the chapters and the main content of each chapter. We hope that this will help clarify the organization and content of the entire thesis.

This thesis is divided into 5 chapters as follows:

Chapter 1: Introduction: In this chapter, we will focus on presenting the context, stating the problem, objectives and importance of this research.

Chapter 2: Literature Review: Presents general knowledge about lung cancer, CT scanning techniques, and applications of deep learning in the medical field. At the same time, we will also present some related studies on this topic.

Chapter 3: Material and Methodology: In this chapter, we will focus on presenting the research steps in this thesis such as data pre-processing, configuration and training of three-dimensional neural network models, and model performance evaluations.

Chapter 4: Results and Discussion: Presents and discusses the results obtained after training the model. From there, we draw out the advantages and disadvantages of this model.

Chapter 5: Conclusion, Limitations and Future Works: In this final chapter, we will give a general summary of the research, the limitations of the research and make recommendations for future research development.

Chapter 2

Literature Review

2.1 Lung Cancer and Screening Methods

2.1.1 Lung Cancer Overview

Lung cancer is an oncological illness and stands high among cancers ranked in terms of how common it is and how many deaths it causes worldwide. In lung cancer patients, abnormal cell proliferation occurs specifically in the lungs where tumors are developed. Over time, these tumors will multiply and become invasive. At that point, they will tend to spread into neighboring structures and eventually spread to other parts of the body, which is called metastasis.

The epidemiology of lung cancer patients has shown that the disease of lung cancer is not one but is multi-factorial in its causes. Alas, the foremost of them has remained the same – tobacco use is still the biggest risk factor for lung cancer today. Cigarette smoke contains harmful chemicals that alter lung cultures and make them divide without regulation. Indirectly inhaling the same smoke that others are exhaling, or secondhand smoke also helps to elevate the risk of lung cancer. Another risk factor that should not be overlooked is the long-term presence of polluted air, particularly fine powders like those found in urban environments nowadays. Other less frequently encountered risk factors such as radon (a naturally inert gas that is found in rocks and soil and that is radioactive), asbestos, and internal family and relatives history, etc. all can contribute to lung diseases.

According to statistics from the World Health Organization (WHO), lung cancer is the leading cause of cancer death globally, accounting for about 1.8 million deaths each year. And there are an estimated 2.21 million new cases of lung cancer each year. In Vietnam, lung cancer is the second most common cancer after liver cancer, especially in men. According to Globocan 2020, there are an estimated 26,351 new cases and 23,515 deaths from lung cancer each year in Vietnam.[1][2]

2.1.2 Current Screening Methods

Currently, we can detect lung cancer through several methods such as chest X-ray, bronchoscopy, sputum test, Magnetic Resonance Imaging (MRI) scan, Positron Emission Tomography (PET) scan, computed tomography (CT) scan, or some other testing methods. Chest X-ray is the basic method and is often the first to be applied to examine the lungs. This method can detect large tumors or other abnormalities in the lungs, but it can miss small tumors or tumors that are difficult to see. Bronchoscopy uses a flexible, flexible tube to look inside the airways and lungs, allowing the doctor to take tissue samples to check for cancer. Another method is computed tomography, a CT scan that can provide more detailed images of the lungs than an X-ray, allowing the detection of small tumors and other abnormalities.

2.2 CT Scan Technology Overview

Computed Tomography scan (CT) is a more sophisticated form of imaging technique involving use of X-ray and a computer to produce pictures of the internal structure of the body. The patient lies on the couch and the couch moves through the CT machine. An X-ray tube rotates around the body and takes many pictures from different positions. The images are then compiled by a computer to develop a more elaborate three-dimensional picture of the lungs and adjacent organs. CT scans play an important role in diagnosing lung cancer, especially in:

Early detection: Low-Dose Computed Tomography (LDCT) enables the identification of small nodules in the lungs that are 3 or 4 millimeters in size, nodules that cannot be diagnosed by conventional chest X-rays. This allows for early detection of lung cancer when the tumor is small and has not yet metastasized thus improving the likelihood of treatment.[3]

Accurate diagnosis: CT scans, therefore help the doctor in the identification of the precise position, size, shape, and nature of the tumor, whether it is benign or malignant, and any encroachment of the tumor to neighboring tissues. CT scans also aid in determining the precise location of the tumor so that an accurate biopsy may be taken to arrive at the diagnosis of cancer.

Staging: Knowing the extent or stage of lung cancer is very important in management, and is well illustrated by a CT scan that can reveal whether cancer has spread to the lymph nodes or any other part of the body. This information is crucial for determining at what phase of the disease the patient is located and which treatment should be used for the patient.

Treatment monitoring: CT scans are employed in the evaluation of the effectiveness of lung cancer treatment with the help of comparisons in variations in the size of tumors following surgery, chemotherapy, or radiation therapy.

That is why CT scans can be considered an effective tool when diagnosing and analyzing lung cancer. However, the interpretation of images obtained from CT

scans presents a lot of difficulty to the physicians particularly where they have a giant pile of work to go through. This has led to the emergence of high demand for the automation of the emanating medical images, especially for physicians to read the CT scan images much faster and accurately.

2.3 Application in Medical Imaging

2.3.1 Application of Deep Learning in Medical Imaging

Deep learning models are advanced artificial intelligence tools that could change how medical diagnosis is done by mimicking the way humans learn. These models can handle large amounts of data and spot complex patterns, making them useful for analyzing medical images and helping doctors make faster and more accurate diagnoses.

Unlike older methods for processing images, deep learning, especially using neural networks, can closely examine medical images like X-rays, CT scans, and Magnetic Resonance Imaging (MRI). They can find small changes, like tumors or other issues, that doctors might miss. For example, a recent study showed that a deep learning model could detect breast tumors in mammograms just as well as experienced radiologists.

Beyond detecting problems, deep learning can help diagnose diseases more accurately and predict a patient's health risks, which can guide doctors in choosing the right treatments. In another study, deep learning was used to classify heart conditions from electrocardiogram (ECG) results with about 98% accuracy. It can even help adjust treatments to better suit individual patients.

2.3.2 Application of 3D CNN in Medical Imaging

A three-dimensional convolutional neural network is a newer type of deep learning network designed to work with three-dimensional data, like CT scans and MRI images. Unlike older neural networks that only look at two-dimensional data, this type of network can also analyze images that have information in three dimensions, which gives better details for diagnosing diseases.

Medical image analysis is an area where 3D CNN have been used a lot and have had some impressive successes. One of the most common techniques is image segmentation, which means cutting out parts of the image, like body structures or damaged areas. For instance, 3D CNN has been utilized for the segmentation of brain tumors in MRI images, thereby enabling the accurate localization and dimension appraisal of the tumors for surgical procedures. This helps in alleviating problems of clinical disease assessment and helps the doctors in computing the disease's urgency and seriousness indicators. At the same time, 3D CNN is also employed in detecting pathological changes in the internal organs of patients by evaluating medical images, for instance, tumors, lesions, or other pathology, with high efficiency

and accuracy. For instance, in the study, 3D CNN was employed to perform highly accurate detection of malignant lung nodules on CT from healthy subjects, with high sensitivity. Additionally, by understanding the structure of diseases through medical images, 3D CNN can also assist doctors in making treatment decisions.

2.4 Related Works

CNN based on DL has recently shown up as a very powerful technique to analyze medical images, especially for the diagnosis of lung cancer using CT scans. Some papers investigated the use of 3D CNNs in the context of lung cancer diagnosis showing that they might outperform standard approaches.

For example, one research by Setio et al. (2016) introduced a multi-view convolutional to help with automatic pulmonary nodule detection in CT images. Their strategy was based on training 3D CNNs on different views of the input data enlarging thus the information coming from multiple planes and increasing the sensitivity of nodule detection. This demonstrates the usefulness of 3D CNNs for the extraction of feature representation from volumetric medical image data.[4]

Anthimopoulos et al. (2016) also conducted another study to explore the deep convolutional neural network owing to its potential for lung pattern classification of interstitial lung diseases. Their efforts showed that 2D CNNs can accurately classify lung patterns and hence CNNs hold a future for automated diagnosis of different lung diseases. Although their aim was 2D-based it emphasize the extendability of CNNs in pulmonary image analysis.[5]

Considering lung cancer diagnosis, Jiang et al. (2019) suggested the use of the 3D CNN for the classification of lung nodules. Local information was obtained through feature extraction at various scales, based on their multi-scale model. This approach improved the classification accuracy and hence supports the analysis that the multi-scale analysis in 3D CNNs is important in the diagnosis of lung cancer.[6]

In addition, the work done by Nibali et al. (2017) examined the possibilities of 3D CNNs for lung nodule malignancy prediction. They found that augmented data along with transfer learning gives enhanced results while working with 3D CNNs when there is a limited amount of training data.[7]

Chapter 3

Material and Methodology

3.1 Material

The dataset used in this study is a subset of the publicly available dataset provided by the National Cancer Institute (NCI) and the National Lung Screening Trial (NLST) collaboration. National Cancer Institute is a leading cancer research organization in the United States, and the National Lung Screening Trial is a large-scale clinical trial sponsored by NCI to determine the effectiveness of lung cancer screening using CT scans. The primary source of data comes from large clinical trials involving lung cancer screening conducted by these prestigious medical institutions.

The original dataset consists of low-dose CT scans of 3,382 patients, which are divided into a training dataset of 1,397 patients, each with a set of CT scans of the lungs and a corresponding label indicating whether the patient has lung cancer or not. The test dataset consists of 1985 patients, this test dataset only contains CT scan images without providing labels. However, as mentioned above, the dataset used in this study is just a subset extracted from the original dataset which consists of 769 patients only.[8]

However, there are a few issues with this dataset such as the difference in the number of CT scan slices of each patient, and the imbalance in the dataset with the number of patients without lung cancer being the majority ($>70\%$). Below are some statistics of the above dataset:

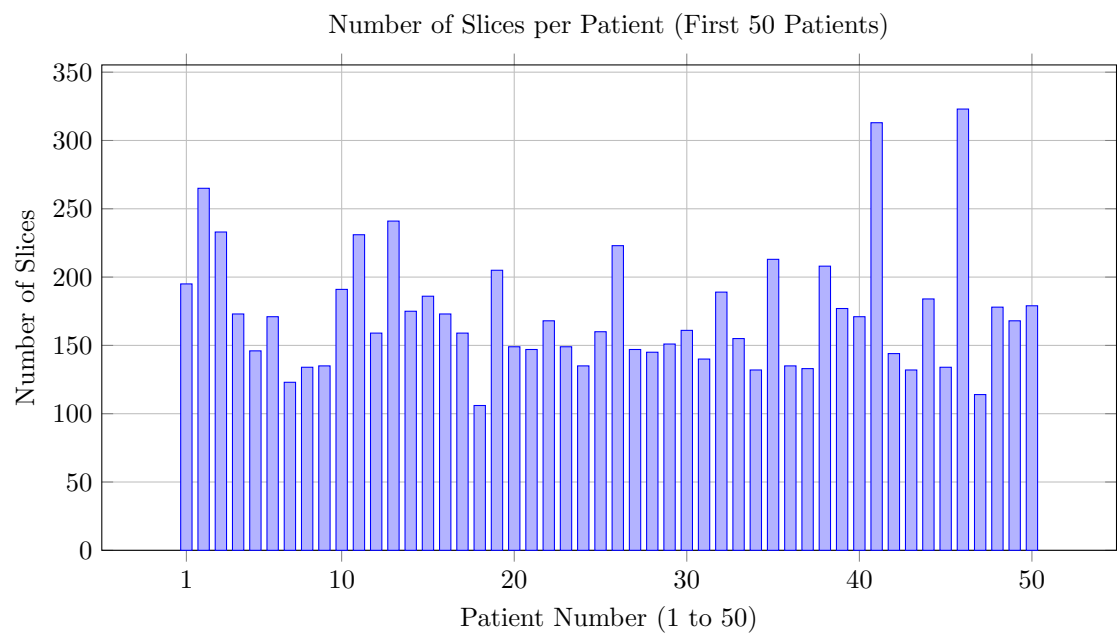


Figure 3.1: Number of slices per patient

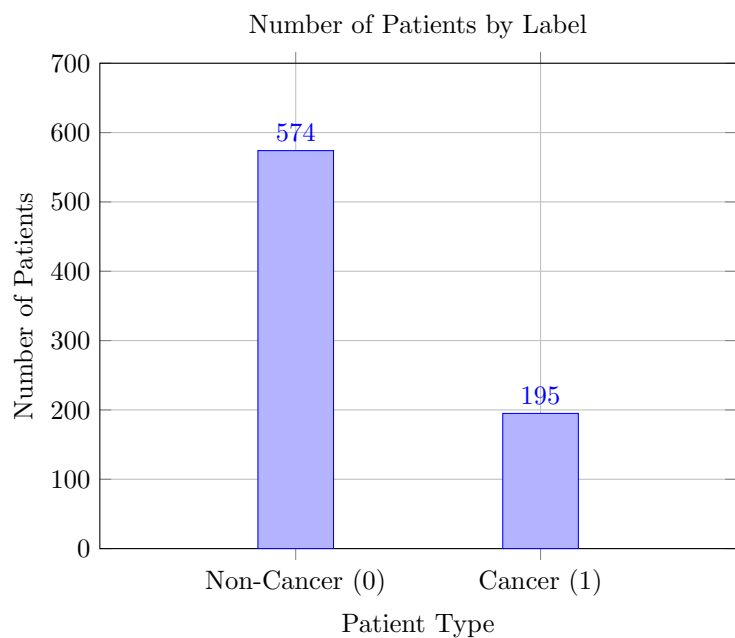


Figure 3.2: Number of patients by label

To get a more visual perspective on this dataset, here are some visualizations of the dataset:



Figure 3.3: Lung CT scans

Low-dose CT scans of patients are stored in the .dcm or DICOM format, which stands for Digital Imaging and Communications in Medicine. This format is an international standard for storing and processing medical information, such as X-rays, MRI scans, or in this case, low-dose CT scans. Here are some of the key attributes of the DICOM format, taken as an example of a random patient in the dataset:

Table 3.1: Important attributes used in the study

Attribute	Value	Description
ImagePositionPatient	[25.4, -120.8, 62.3] mm	(x, y, z) coordinates of the slice in 3D space
SliceLocation	150.5 mm	Slice location, used to calculate slice thickness
SliceThickness	2.5 mm	Thickness of each slice
PixelSpacing	[0.7, 0.7] mm	Size of each pixel (in 2D plane)
Rows	512	Number of rows in the image
Columns	512	Number of columns in the image
RescaleIntercept	-1024	Intercept value for converting to Hounsfield Units.
RescaleSlope	1	Slope value for converting to Hounsfield Units

In the provided dataset, we also have information about the labels of each patient. The label file is saved in CSV format and includes the ID of each patient and the label corresponding to each patient (0: no cancer and 1: cancer). Here are some examples:

Table 3.2: A sample of the labels.csv file

Patient Id	Label
0015ceb851d7251b8f399e39779d1e7d	1
0030a160d58723ff36d73f41b170ec21	0
003f41c78e6acfa92430a057ac0b306e	0
006b96310a37b36cccb2ab48d10b49a3	1
008464bb8521d09a42985dd8add3d0d2	1
0092c13f9e00a3717fdc940641f00015	0
00986bec45e12038ef0ce3e9962b51a	0
00cba091fa4ad62cc3200a657aeb957e	0
00edff4f51a893d80dae2d42a7f45ad1	1
0121c2845f2b7df060945b072b2515d7	0
01de8323fa065a8963533c4a86f2f6c1	0
01e349d34c06410e1da273add27be25c	0
01f1140c8e951e2a921b61c9a7e782c2	0
024efb7a1e67dc820eb61cbdaa090166	0
0257df465d9e4150adef13303433ff1e	1

3.2 Methodology

3.2.1 Overall Framework

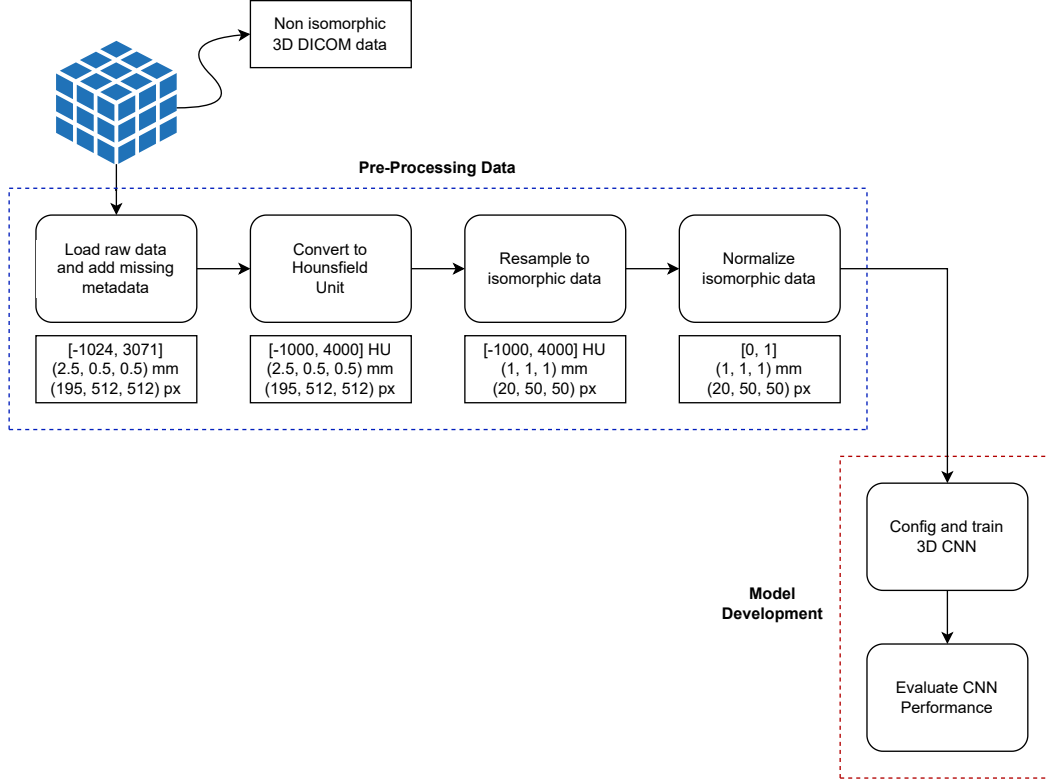


Figure 3.4: Overall framework

In any research, having a clear and coherent methodological framework is essential to ensure the consistency, logic and effectiveness of the research process. This framework acts as a road map, outlining the steps required to go from defining the research problem to drawing conclusions and discussing their implications. The diagram above gives an overview of the processes that will be presented in this project.

3.2.2 Pre-Processing Data

The dataset used in this study is stored in DICOM format, a standard file format commonly used in medical imaging because it contains not only image data but also a lot of metadata such as patient information, imaging equipment, and other technical parameters. However, these raw data are very complex and have not been converted to a standard form ready to be trained by deep learning models. Therefore, we must pre-process these raw CT scan data to clean and standardize the data before training the model. This increases the efficiency of model training and helps improve the accuracy of deep learning models. In this study, we have performed data cleaning and standardization steps such as loading and sorting DICOM data files, converting

to standard HU units, resampling, and pixel normalization. Here we will explain in detail the data pre-processing steps performed in this study:

Step 1: Load data and add missing metadata:

In the first step, we used the `load_scan()` function to read the patient's CT scan data. Here we consider each patient to correspond to a CT scan image and each CT scan image is made up of many slices, each slice corresponding to a DICOM file.

After reading the patient's CT scan data, we found that the DICOM slices were not arranged in their correct order. Arranging the DICOM files in the correct order is very important because we can ensure that the data is processed in the correct anatomical order, and later we can also use it to create a complete 3D scan of the lung.

We can easily rearrange the lung slices based on the information stored in the `ImagePositionPatient` field. This information can be easily extracted and manipulated by accessing the metadata of each DICOM file.

Another issue that needs to be noted is that the metadata of DICOM files lacks information about slice thickness or pixel size in the Z direction, and we are only provided with information about pixel size in the X and Y directions. Here, this information tells us about the size of each pixel or more precisely, the volume of each voxel in 3D space.

We can handle this problem by calculating the thickness parameter of each slice by calculating the distance between two consecutive slices from the `ImagePositionPatient` information. From there, we can add information about the `SliceThickness` field to the metadata of the DICOM files so that we can use them in the next steps. Here we have integrated reading data from DICOM files, reordering the slices in the correct anatomical order, calculating the thickness between slices, and adding them to the file metadata into the `load_scan()` function. After going through the above steps we will get information about the `SliceSpacing` field of the patients as [2.5, 0.5, 0.5], which means we can now know the volume of a voxel in 3D space is 2.5 mm x 0.5 mm x 0.5 mm.

From the above idea, the proposed pseudocode had been used for step 1:

Algorithm 1 The proposed pseudocode for load data and add missing metadata

Require: *path* - The path to the folder containing DICOM files

Ensure: *slices* - A list of DICOM slices sorted by their Z position

- 1: *slices* = `READ_DICOM_FILES(path)`
 - 2: *slices* = `SORT_SLICES_BY_Z_POSITION(slices)`
 - 3: *slice_thickness* = `CALCULATE_SLICE_THICKNESS(slices)`
 - 4: `ADD_SLICE_THICKNESS_TO_METADATA(slices, slice_thickness)`
 - 5: **return** *slices*
-

Step 2: Convert raw pixel to Hounsfield Units (HU):

The Hounsfield scale is a scale used to measure the X-ray attenuation density of tissues in the body; it is widely used in computed tomography. The Hounsfield scale works based on the principle of X-ray absorption of tissues, because each type of tissue in the body will have a different level of X-ray absorption, so from the results returned from the Hounsfield scale we can determine the tissues in the body in detail.

The formula for Hounsfield Units (HU) is given by:

$$\text{HU} = 1000 \times \frac{\mu_{\text{voxel}} - \mu_{\text{water}}}{\mu_{\text{water}} - \mu_{\text{air}}}$$

Where:

μ_{voxel} : X-ray attenuation coefficient of the voxel.

μ_{water} : X-ray attenuation coefficient of water.

μ_{air} : X-ray attenuation coefficient of air.

After converting raw pixel values to HU units, we can use the following reference table to classify different tissues:

Table 3.3: Substances and their corresponding Hounsfield Units

Substance	Hounsfield Unit
Air	-1000
Fat	-100 to -50
Water	0
Muscle	40 to 60
Liver	50 to 70
Bone (Cortical)	700 to 3000
Blood	40 to 60
Soft Tissue	30 to 100
Calcium	300 to 1000
Lung	-500 to -700
Contrast Agent	Varies (typically > 100)

Raw pixel values in DICOM images often depend on many objective factors such as scanner models, specific technical settings of each machine, etc. This leads to differences in pixel intensity between different captures even when they are used with the same type of machine model. Therefore, we need to normalize pixel data by converting it to Hounsfield units (HU) to eliminate this difference. This will bring pixel values to the same scale and thereby enhance the analysis ability and improve the performance of the model later.

In DICOM images, raw pixel values are often stored as unsigned integers, so they cannot reflect the density of tissues. To convert to Hounsfield units, we will use parameters such as `RescaleIntercept` and `RescaleSlope` taken from the metadata

of the DICOM file. Thus, we use the conversion formula:

$$\text{HU} = \text{RescaleSlope} \times \text{pixel_value} + \text{RescaleIntercept}$$

For example, a voxel has a raw pixel value of 1000, and with the parameters $\text{RescaleIntercept} = -1024$, $\text{RescaleSlope} = 1$. After converting to HU, we will get the new value of this voxel:

$$\text{HU} = 1 \times 1000 + (-1024) = -24$$

In this step, we have written the `get_pixels_hu()` function to perform this unit conversion; the value returned by the function will be saved in a 3-dimensional NumPy array format for convenient use in the next steps. Through the pixel values in HU units, we can visualize the tissues in the image.

The proposed pseudocode had been used for step 2:

Algorithm 2 The proposed pseudocode for convert raw pixel to HU

Require: *slices* - A list of DICOM slices

Ensure: *image* - A NumPy array containing pixel values in HU

```

1: image = STACK_PIXEL_ARRAYS(slices)
2: CONVERT_TO_INT16(image)
3: REPLACE_VALUE(image, -2000, 0)
4: for each slice in slices do
5:   intercept = GET_INTERCEPT(slice)
6:   slope = GET_SLOPE(slice)
7:   if slope  $\neq$  1 then
8:     CONVERT_TO_FLOAT64(slice)
9:     MULTIPLY_BY_SLOPE(slice, slope)
10:    CONVERT_TO_INT16(slice)
11:   end if
12:   ADD_INTERCEPT(slice, intercept)
13: end for
14: return image

```

Step 3: Resample

After the CT scan data has been converted to Hounsfield units, we encounter another problem, which is the lack of data uniformity. The lack of uniformity of the data set here is shown by the difference in the number of slices between patients and the uneven image size between slices. This problem can be caused by many reasons, such as the use of different scanner models or different adjusted scanning specifications, ... these can all lead to the creation of images with different resolutions, pixel sizes, and slice thicknesses. Objective reasons such as differences in the patient's body, posture, and position when scanning can also cause differences in the data. Therefore, resampling is an extremely important step before we can use the data to train deep learning models, which often require the data to have a uniform structure.

Resampling is resizing the 3D image data. We do this by using the `resample()` function, which uses the nearest interpolation method `mode='nearest'` to calculate

new pixel values. This method is a fairly common way to resample data and resize images. The way it works is also relatively simple, first, we will determine the new pixel location, then find the pixel in the original image that is closest to the new pixel location and assign a value. By using this method we will be able to resample the data and resize the image.

Our goal when building this `resample()` function is to bring the original size of the voxels in 3D space to 1mm x 1mm x 1mm, at the same time we also limit the number of slices in each patient to 20 slices and resize each slice to 50 x 50 pixels. This helps reduce the size of the data and ensures that the structure of the data is uniform before using it to train deep learning models.

The proposed pseudocode had been used for step 3:

Algorithm 3 The proposed pseudocode for resample

Require:

- 1: *image* - A 3D NumPy array containing pixel data in Hounsfield Units
- 2: *scan* - A list of DICOM slices
- 3: *new_spacing* - Desired voxel spacing in mm (default: [1, 1, 1])
- 4: *new_size* - Desired size of each slice in pixels (default: (50, 50))
- 5: *slice_limit* - Maximum number of slices (default: 20)

Ensure:

- 6: *resized_image* - Resampled image with desired spacing and dimensions
 - 7: *new_spacing* - Actual voxel spacing after resampling
 - 8: *pixel_spacing* = GET_PIXEL_SPACING(*scan*)
 - 9: *spacing* = COMBINE_SPACING(*scan*, *pixel_spacing*)
 - 10: *resize_factor* = CALCULATE_RESIZE_FACTOR(*spacing*, *new_spacing*)
 - 11: *new_shape* = CALCULATE_NEW_SHAPE(*image*, *resize_factor*)
 - 12: *real_resize_factor* = CALCULATE_REAL_RESIZE_FACTOR(*new_shape*, *image*)
 - 13: *new_spacing* = CALCULATE_NEW_SPACING(*spacing*, *real_resize_factor*)
 - 14: *image* = RESAMPLE_IMAGE(*image*, *real_resize_factor*)
 - 15: *image* = LIMIT_SLICES(*image*, *slice_limit*)
 - 16: *resized_image* = RESIZE_SLICES(*image*, *new_size*)
 - 17: **return** *resized_image*, *new_spacing*
-

Step 4: Normalization

After converting the raw pixel values to Hounsfield units and re-sampling the entire dataset, we found another problem: the range of CT scan pixel values is still relatively wide. The pixel values in the Hounsfield scale now range from -1000 HU to 4000 HU. With such a wide range, this can create problems during model training, such as the model may have difficulty converging or the results may be unstable. When the pixel values have such a wide range, features with large values will dominate those with small values during training, making it difficult for our model to learn the important features.

Therefore, in this step, we will perform data normalization by applying methods such as min-max normalization and zero centering. Applying these methods will help us bring the pixel value range from -1000 HU to 4000 HU to the range [0, 1], and then move the data center to 0.

In this step, we have performed normalization through two functions: `normalize()`

and `zero_center()`. The purpose of the `normalize()` function is to bring the pixel values to the range $[0, 1]$ using the min-max normalization method with the following logic:

$$\text{normalize_pixel} = \frac{\text{pixel} - \text{MIN_BOUND}}{\text{MAX_BOUND} - \text{MIN_BOUND}}$$

Where:

`pixel`: the value of the pixel in Hounsfield units

`MIN_BOUND`: the smallest pixel value in the data set in Hounsfield units

`MAX_BOUND`: the largest pixel value in the data set in Hounsfield units

The purpose of the `zero_center()` function is to move the center of the data to 0. This function subtracts the average pixel value `PIXEL_MEAN` from each normalized pixel in the range $[0, 1]$:

$$\text{Zero_centered_pixel} = \text{normalized_pixel} - \text{PIXEL_MEAN}$$

The proposed pseudocode had been used for step 4:

Algorithm 4 The proposed pseudocode for normalization

Require: *image* - A 3D NumPy array containing pixel data in Hounsfield Units

Ensure: *normalized_image* - A 3D NumPy array with pixel values normalized and zero-centered

- 1: *normalized_image* = `NORMALIZE(image)`
 - 2: *normalized_image* = `ZERO_CENTER(normalized_image)`
 - 3: **return** *normalized_image*
-

3.2.3 Model Configuration and Training

After performing the data pre-processing steps, the next step is to configure and train the deep learning model to classify lung cancer patients or non-cancer patients. The model we use here is a 3D CNN model, the reason why we choose this model is not special. We think that we have chosen to use the 3D CNN model in this study because of the naturalness of the data. Next, we will present the details of the architecture of the 3D CNN model used, the hyper-parameters and the model training process. Here is the architecture of 3D CNN model:

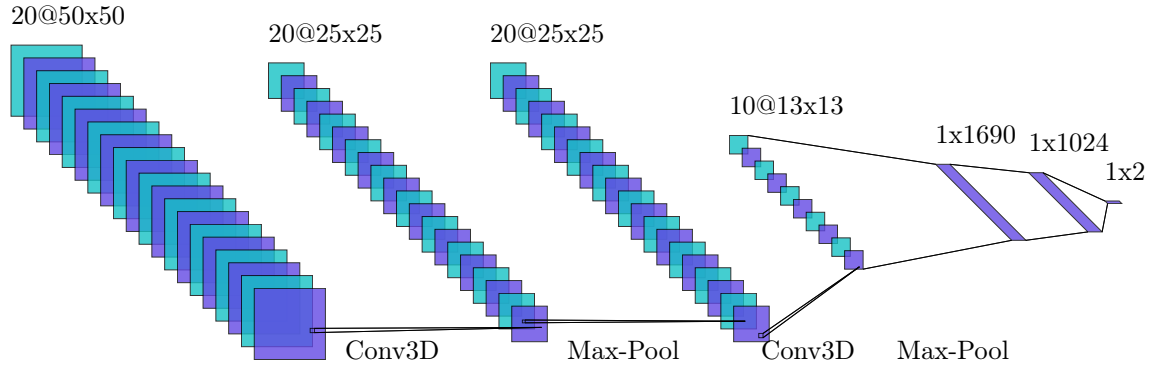


Figure 3.5: 3D CNN model architecture

Table 3.4: 3D CNN model architecture

Layer	Description
Convolutional 1	This layer is used to extract basic features of input data. This layer consists of 32 filters with kernel size (3,3,3), ReLU activation function and 'same' padding
Max Pooling 1	This layer reduces the size of the feature map, which reduces the number of parameters and increases the model's invariance to small changes in the data. Pool size (2,2,2) and padding 'same'
Convolutional 2	This layer is used to extract more complex features from the feature map of the previous layer. This layer consists of 64 filters with kernel size (3,3,3)
Max Pooling 2	Similar to Max Pooling layer 1, this layer also reduces the size of the feature map. Pool size (2,2,2) and padding 'same'
Flatten	This layer converts the output of the previous layer into a 1-dimensional vector that can be fed into the Fully Connected layer
Fully Connected	This layer consists of 1024 neurons with ReLU activation function used to learn complex relationships between features
Dropout	This layer prevents overfitting by randomly dropping some neurons during model training process with a dropout rate of 0.2
Fully Connected (Output)	This layer gives the predicted probability for each class, and this layer consists of 2 neurons corresponding to 2 classes: cancer and non-cancer classes, the activation function used here is Softmax function

Hyper-parameters:

Table 3.5: Hyper-parameters

Hyper-parameter	Value	Description
IMG_SIZE_PX	50	Size of each slice after resample
SLICE_COUNT	20	Number of slices in each 3D data block
n_classes	2	Number of classes (0 means non-cancer and 1 means cancer)
batch_size	10	Number of samples used in each weight update
keep_rate	0.8	Percentage of neurons retained in the Dropout layer
hm_epochs	10	Number of epochs (iterations through the entire data)

Training the 3D CNN Model:

In this section, we will present the details of the training process of a 3D CNN model with the configuration described above. First, we will load the processed data, which includes CT scans of patients stored as NumPy arrays, and the corresponding labels of those patients. Next, we will use the `tf.keras.utils.to_categorical` function to convert the labels of the patients to one-hot encoding. For example, a patient with a label of 0 will be converted to `[1, 0]`, while a label of 1 will be converted to `[0, 1]` since the output of the model is a probability vector. Then, the data will be divided into a training dataset and a testing dataset with a ratio of 9:1. The data will be reshaped into the form `(batch_size, IMG_SIZE_PX, IMG_SIZE_PX, SLICE_COUNT, 1)` to be able to be fed into the 3D CNN model. Here we just need to note that the last parameter represents the number of color channels, however in this case our CT image data is a gray-scale image so there is only 1 color channel.

The loss function we choose here is the categorical cross-entropy function, which measures the difference between the predicted probability distribution of the model and the actual probability distribution of the label.

We use the Adam algorithm as the optimization algorithm in the process of training this 3D CNN model, the learning rate is set to `1e-3`. This is an optimization algorithm based on the Gradient Descent algorithm and it is a popular optimization algorithm in deep learning.

After completing the data preparation, choosing the appropriate loss function and the appropriate optimization algorithm, we will proceed to train the 3D CNN model using the `model.fit()` function. With the training data, training labels, batch size, number of epochs and test data, this function will update the model weights for us with the purpose of minimizing the loss function.

Finally, after the model training process is finished, we will use the `model.evaluate()` function to evaluate the performance of the 3D CNN model. This function will return the loss and accuracy values of the model on the test dataset. Then we will save the results to visualize the training results of this 3D CNN model. The trained model will be saved to the file `cnn3d_model.h5` and the training history will be saved to the file `training_history.pkl`.

3.2.4 Evaluation Metrics

After training, it is time to evaluate the performance of the 3D CNN model. Evaluating the performance of the model is an extremely important step because, through the evaluation metrics, we can know the ability to classify lung cancer or non-lung cancer based on CT scans. Not only that, the evaluation metrics will also show us the advantages and disadvantages of this model. In this section, we will present the evaluation metrics that have been used to evaluate this 3D CNN model.

(1) Accuracy

Definition: The proportion of correctly classified samples out of the total number of samples.

Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

TP (True Positive): The number of cancer samples that are correctly classified as cancer.

TN (True Negative): The number of non-cancer samples that are correctly classified as non-cancer.

FP (False Positive): The number of non-cancer samples that are misclassified as cancer.

FN (False Negative): The number of cancer samples that are misclassified as non-cancer.

Advantages: Easy to understand, easy to calculate, and commonly used.

Disadvantages: Can be misleading when the dataset is unbalanced (e.g., the number of cancer samples is much smaller than the number of non-cancer samples). In this case, even a simple model that always predicts "non-cancer" can achieve high accuracy. Does not distinguish between types of errors (FP and FN). In lung cancer diagnosis, missing a cancer case (FN) is much more serious than misdiagnosing a non-cancer case as cancer (FP).

(2) Sensitivity

Definition: The proportion of cancer samples that are correctly classified as cancer out of the total number of actual cancer samples.

Formula:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

Advantages: Measures the ability of the model to correctly detect cancer cases. Important in medical applications where missing a cancer case (FN) can have serious consequences.

Disadvantages: Does not consider the number of non-cancer samples that are misclassified (FP).

(3) Specificity

Definition: The proportion of non-cancer samples that are correctly classified as non-cancer out of the total number of actual non-cancer samples.

Formula:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Advantages: Measures the ability of the model to correctly exclude non-cancer cases.

Disadvantages: Does not consider the number of misclassified cancer samples (FN).

(4) Precision

Definition: The proportion of samples classified as cancer that are actually cancer over the total number of samples classified as cancer.

Formula:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Advantages: Important when the cost of misdiagnosing a non-cancer case as cancer (FP) is high.

Disadvantages: Does not consider the number of misclassified cancer samples (FN).

(5) F1-score

Definition: The F1-score is the harmonic mean of precision and recall. It provides a balanced measure of a model's accuracy, especially in situations with imbalanced datasets.

Formula:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Advantages: It considers both false positives and false negatives, providing a more comprehensive evaluation than either metric alone. And useful when dealing with datasets where one class is significantly more prevalent than the other.

Disadvantages: Less intuitive than precision and recall. It can be harder to interpret directly compared to the simpler concepts of precision and recall. And may not be suitable for all applications. In some cases, we might prioritize either precision or recall more heavily depending on the specific problem.

(6) AUC-ROC

Definition: Area under the Receiver Operating Characteristic (AUC-ROC) curve. The ROC curve represents the relationship between True Positive Rate (TPR) (equivalent to sensitivity) and False Positive Rate (FPR) ($FPR = 1 - \text{Specificity}$) at different classification thresholds.

Advantages: Evaluates the performance of the model over the entire range of classification thresholds. Unaffected by data imbalance.

Disadvantages: Difficult to interpret visually for non-experts.

3.2.5 Tools and Libraries

In this study we have used several software tools and libraries to perform various tasks such as reading and processing DICOM image datasets, pre-processing data, building and training 3D CNN models, and visualizing the data. Following is the list of tools and libraries we have used in this study:

Table 3.6: Tools and libraries

Tools and Libraries	Description
Python	This is the main programming language used in this study due to its flexibility, ease of use, and many supporting libraries.
NumPy	The library provides multidimensional array structures and mathematical functions for fast and efficient manipulation of arrays.
Pandas	Library providing algorithms for manipulating tabular data and time series data.
Matplotlib	Library providing algorithms for data visualization.
Sklearn	Library providing machine learning algorithms.
OpenCV	Library providing image processing and computer vision algorithms.
Tensorflow	Open source platform for training machine learning and deep learning models.
Keras	API for Tensorflow simplifies building and training machine learning, deep learning models.
SciPy	Library providing algorithms for scientific computing in general.
pydicom	Library for reading and working with DICOM files.

Chapter 4

Results and Discussion

4.1 Results

Unfortunately but not too surprised with the obtained results. The model is heavily biased towards predicting non-cancer (Class 0) and fails to identify any cancer cases (Class 1) due to the problem of imbalance between classes and low signal-to-noise. As mentioned in the abstract, it's inherently difficult to detect cancer from CT scans, even for humans. The model might be struggling to learn the subtle features that distinguish cancer from non-cancer. In essence, our model is effectively useless for its intended purpose.

Table 4.1: Model performance

Evaluation Metric	Value
Accuracy	0.68
Sensitivity (Recall)	0.0
Specificity	1.0
Precision	1.0
Area Under the ROC Curve (AUC)	0.61
Number of Class 0 (non-cancer) in test set	68
Number of Class 1 (cancer) in test set	32

4.2 Discussion

Analysis of Model Performance:

The accuracy of 0.68 is misleading in this case. Since our dataset contains more non-cancer cases, predicting everything as non-cancer would result in a similar accuracy, close to the proportion of non-cancer cases in the data.

The most concerning metric is sensitivity (or recall), which is a troubling 0.0. This means that the model failed to identify any of the actual cancer cases. Unfortunately, it completely missed all 32 cancer patients.

While the specificity is perfect at 1.0, meaning the model correctly flagged all non-cancer cases, this isn't as useful here. In a situation like this, missing cancer diagnoses is far more serious than incorrectly identifying someone as having cancer.

Precision, also at 1.0, indicates that in every case the model predicted as non-cancer was indeed non-cancer. However, since the model didn't predict any cancer cases at all, this metric doesn't hold much significance.

The AUC of 0.61 suggests the model has some ability to distinguish between cancer and non-cancer cases, but it's still far from ideal. An AUC of 0.5 represents a random chance, and we're only slightly above that, so there's room for improvement.

Proposed Improvements:

The model’s performance, as evidenced by the low accuracy and sensitivity, highlights the need for further refinement and exploration of alternative approaches. The high rate of false negatives is particularly concerning, as it could lead to missed diagnoses and delayed treatment for lung cancer patients.

One potential avenue for improvement is to incorporate lung segmentation as a preprocessing step. By accurately isolating the lung regions from the surrounding anatomy, we can effectively reduce noise and irrelevant information, allowing the model to focus on the areas of interest. This could potentially lead to more accurate and reliable predictions.

Furthermore, data augmentation techniques, such as random cropping, rotation, and flipping, can artificially increase the training dataset’s size and diversity. This can help improve the model’s generalization ability and reduce overfitting, leading to better performance on unseen data.

In addition to these preprocessing and data augmentation strategies, exploring alternative model architectures may also yield significant improvements. While 3D CNNs have shown promise in medical image analysis, other architectures like 3D ResNet or DenseNet may be better suited for this specific task. These models have demonstrated superior performance in various image recognition tasks due to their ability to learn more complex and discriminative features.

By combining these strategies—lung segmentation for ROI extraction, data augmentation for increased training data, and exploration of alternative model architectures—we can potentially overcome the limitations of the current model and develop a more robust and accurate system for lung cancer diagnosis.

Chapter 5

Conclusions, Limitations and Future Works

5.1 Conclusions

The primary purpose of this study was to evaluate the proposed three-dimensional convolutional neural network (3D CNN) for classifying lung cancer from CT scans to aid doctors in diagnosis. Unfortunately, our attempts to train a 3D CNN over full lung images were largely unsuccessful.

This project involved data preprocessing for model construction using 3D CNN, model training, and overall performance review, culminating in the revelation of strengths and weaknesses of the 3D CNN model in classifying lung cancer from CT scan images.

Our work has shown that while medical imagery may be amenable to productive analysis and prediction using computer vision techniques, developing models that work well on medical data is not straightforward. We encountered several challenges with medical imagery, including the lung cancer datasets we worked with, that make algorithm-based classification and detection particularly tricky.

Specifically, the limited success of our 3D CNN model can be attributed to the following factors:

Lack of lung segmentation and ROI extraction: The model was trained on full lung images without prior segmentation to isolate the regions of interest (ROIs). This likely introduced significant noise and irrelevant information, hindering the model’s ability to learn discriminative features for accurate classification.

Limited data augmentation: The study did not utilize data augmentation techniques to artificially increase the size and diversity of the training dataset. This limited the model’s ability to generalize and potentially led to overfitting, resulting in poor performance on unseen data.

Suboptimal model architecture and hyperparameters: The specific 3D CNN architecture and hyperparameters used in the study may not have been optimal for this task. Further exploration of alternative architectures and more extensive hyperparameter tuning could potentially yield improved performance.

Despite these limitations, this study provides valuable insights into the challenges and considerations for developing deep learning models for lung cancer classification from CT scans. Future research should prioritize incorporating lung segmentation, data augmentation, and exploring more advanced model architectures to improve classification accuracy and clinical utility.

5.2 Limitations

Despite the progress in the project, some limitations still exist:

Data: The data in the study is very small in number and involves a small group of patients. This does pose some questions about the generality of the model when applied to a larger population of patients. Moreover, the data collected may be imbalanced in terms of classes, which creates the problem of bias when classifying the results. Additionally, the model was not validated on an independent external dataset, which is crucial for assessing its robustness and generalizability.

Model: The used 3D CNN model is highly prone to over-fitting, which is conditional on the volume of training data. This may also lower the efficiency of the model when it is faced with data that had not been seen by the program before. Moreover, the efficacy of early detection of some types of lung cancer may be limited, particularly for those with subtle or less distinct features.

Method: Pre-processing steps may not be comprehensive enough to address all potential variations and artifacts in CT scans, potentially affecting the model’s performance. The hyper-parameters optimization may not have been fully optimized, potentially leaving room for performance improvement.

Resources: The study was conducted under time constraints, potentially limiting the extent of model exploration and optimization. Access to more powerful computational resources could enable the exploration of larger models and more extensive hyperparameter tuning. Limited domain expertise in medical imaging and lung cancer could potentially affect the interpretation of results and the development of clinically relevant solutions.

5.3 Future Works

Based on the results and limitations of this study, several potential directions for future research emerge:

Lung Segmentation: Implement lung segmentation techniques to extract the regions of interest (ROIs) from the CT scans. This will help reduce noise and focus the model’s attention on the relevant anatomical structures.

Data Augmentation: Employ data augmentation techniques, such as random cropping, rotation, and flipping, to artificially increase the size and diversity of the training dataset. This can help improve the model’s ability to generalize and reduce overfitting.

Advanced Architectures: Explore more advanced 3D CNN architectures, such as 3D ResNet or DenseNet, which have demonstrated superior performance in various image recognition tasks.

Hyperparameter Tuning: Conduct a more thorough hyperparameter search to optimize the model’s configuration and improve its performance.

Lung Cancer Subtypes: Investigate the classification of different lung cancer subtypes to provide more specific and targeted diagnoses.

Cancer Stage Prediction: Develop models capable of predicting the stage of lung cancer, which is crucial for treatment planning and prognosis.

Tumor Segmentation: Explore techniques for accurate tumor segmentation within CT images, which can aid in surgical planning and radiation therapy.

Comprehensive Diagnostic System: Work towards building a comprehensive lung cancer diagnosis support system that integrates various imaging features and clinical data to assist physicians in making informed decisions.

Clinical Applications: Evaluate the potential applications of the improved models in clinical settings, including lung cancer screening, diagnosis, treatment planning, and monitoring.

By pursuing these future directions, we can potentially overcome the limitations of the current study and develop more robust and accurate tools for lung cancer detection and management.

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